



MIM Prep Algo & Test-Taking Strategies Sheet

I. How the GMAT Algorithm Works

The GMAT is a computerized adaptive test (CAT), meaning it will choose problems according to your performance on preceding questions. The test begins with randomized problems; once it has accumulated a meaningful sample of responses, it will assign subsequent problems adaptively - increasing the overall difficulty if you are answering most problems correctly and decreasing it if you are answering most problems incorrectly. The questions you receive are also subject to further restrictions - for instance, each test-taker must receive the same balance of topics and question types - resulting in an extremely complicated selection algorithm. Finally, each section of the test will contain 5 - 10 experimental questions (problems being calibrated before they are included in future exam administrations), which are distributed at random and affect neither your score nor your adaptive performance.

In general, you shouldn't worry about the exact difficulty level of problems, or about the nuances of the testing algorithm. The test won't show you the difficulty levels of questions, nor will you be able to guess those levels accurately. Even if you could guess difficulty levels, you still wouldn't be able to identify experimental questions, which could give a completely misleading impression of your performance. For instance, a strangely easy problem could mean that you have bombed the last few problems but is more likely just a random experimental question. You should therefore focus on how adaptive testing changes your strategy and perspective. For instance:

- Because the test chooses questions according to your previous performance, you cannot skip any question, leave any question blank, or return to any previous question. As a result, time management is much more important than on a paper test, because you cannot see any other question until you have answered the current one. Getting hung up on even a single question can have disastrous consequences!
- There is very little correlation between your score and the number of questions you have answered correctly (except at the extremes of the scoring scale). It is possible for a 490 scorer and a 720 scorer to miss the same number of problems!

Common Questions we had when passing the GMAT:

1. Is the entire GMAT adaptive?

Almost all large-scale standardized tests contain some number of “experimental” or “pretest” questions that are administered to the student but do not count toward the student's final score. This is simply a way for the test makers to gather data on the questions, to determine how difficult they are and how well they distinguish between students at different ability levels. They also use the data collected to identify bad questions, so that they can eliminate or fix them before they count.



Some tests, like the LSAT, include all of the pretest questions in a single section. Others, like the GMAT, intermingle the pretest questions with the operational ones. Which section is the pretest section, and which questions are the pretest questions, is usually a well-guarded secret. It is a generally bad strategy to spend time trying to guess whether a given question is operational or not. The price of guessing incorrectly is just too high.

2. **How does the GMAT select which questions I get?**

CATs like the GMAT have a blueprint - a set of specifications (difficulty, question type, content area, etc.) that define which questions you see. At the same time, each question has certain statistical characteristics that the algorithm uses, based on your response, to estimate your quantitative or verbal ability. The algorithm looks at your performance on the questions you have already answered and the characteristics of each question remaining in the pool and then selects for you the question that simultaneously best satisfies the blueprint and provides the most statistical information it can, to generate the best estimate of your ability.

3. **My score doesn't seem to match my performance: I only got a few questions wrong, but my score isn't as high as I thought it would be / I got a bunch of questions wrong, yet my score seems higher than it should be.**

Most exams are linear assessments, like the SAT or your 10th-grade history final. These are scored by counting the number of questions you answer correctly, and sometimes by penalizing for each question you answer incorrectly. The result, a raw score, is then converted to a scaled score, like the 600-2400 range for the SAT.

A computer-adaptive test (CAT) works very differently. It doesn't care as much about how many you get right or wrong, but **rather which questions you get right and wrong**. The CAT algorithm estimates your ability based on a variety of criteria, including the difficulty of a question. After each question, it evaluates your response and updates this estimate. When the test is over, the algorithm converts your quantitative and verbal ability estimates into the quantitative and verbal scaled scores, and then separately combines your quantitative and verbal ability estimates to calculate the overall score.

4. **Do the first X number of questions matter more?**

Many variables come into play when the CAT selects your next question. One of them is the CAT's current estimate of your ability. It uses this estimate to select questions that will be most useful in refining that estimate (if you're a high-performing student, giving you low-difficulty questions isn't usually as useful in discerning your true ability as giving you harder questions, and vice versa). What is important to remember is that you should not try to guess how you are doing by whether the question in front of you seems easy or difficult; every question deserves your full attention. With that understood, unless you have completely bombed the test, it is



usually the case that missing a couple of very hard questions late in the test will have a smaller effect on your final score than missing a couple of very easy questions earlier, not because of their position within the test but because of their levels of difficulty.

5. **How severe is the penalty for not finishing a section?**

The penalty is significant. You can expect your scaled score to decrease by roughly 1 point for every question that you don't answer. For example, if you correctly answer every question, you encounter but fail to answer the last five, you generally won't score higher than a 46.

6. **I took the GMAT and got a 710, 44Q/44V/6AWA. A friend of mine took the test 6 days later and got the same quant/verbal scaled scores but he got a 720. How could this happen?**

Both the individual section scores and the overall score are calculated using an estimate of your Math and Verbal abilities derived from your performance on the CAT. Your overall score is not calculated from your section scores. Because your underlying ability estimate might be slightly different from your friend's, your overall scores might be different.

For example, there is a range of ability estimates that translate into a Verbal score of 40, and there is a range of ability estimates that translate into a Math score of 42. Depending on which specific estimate is calculated for you, your overall score could range from 660 to 680. Please note that the Standard Error of Measurement (SEM) on the overall score for GMAT is 29 points, so scores of 660 / 680 all fall within the standard error.

7. **How can my overall percentile be higher than both my quantitative and verbal percentiles?**

Your overall score is calculated separately from your section scores, so you can score in the 99th percentile on the GMAT even if you didn't score in the 99th percentile on either of the sections. For example, you could get a 48 on Quantitative (86th percentile), a 45 on Verbal (98th percentile), and a 760 overall (99th percentile).

8. **Are the quantitative and verbal sections weighted equally in the total score?**

Technically, yes - the estimates of your quantitative and verbal abilities that the CAT produces contribute the same amount to your overall score. However, the verbal section has a greater effect on your percentile rank because it is generally more difficult. If, for example, you scored a 40 on both the Quantitative and Verbal sections, your percentile rank for Quantitative would be 61st, but for Verbal it would be 91st. Your overall score (650) would be in the 84th percentile.



II. The Golden/Cardinal Rules for Passing the GMAT

1. $Q1 > Q2 > Q3 > Q4$

Understanding how GMAT quartiles work is pivotal to maximising your GMAT score. Candidates tend to believe all questions are weighted relatively equally and thus should pass exactly the same amount of time on each question. **This could not be further from the truth.**

How you perform on each quartile (i.e quarter of questions) will play a determining role on your overall performance - the GMAT algorithm weighs very heavily your accuracy on Q1, heavily on Q2, and subsequently less heavily on Q3 and Q4.

What does this mean? **This means that you can have a better GMAT score with more errors if and only if more/most of your errors are located in Q3 and Q4**

2. Limit to the best of your ability your amount of consecutive errors

Another very important metric taken into account by the GMAT algorithm is **how close apart your errors are from one another** - in this regard, **series of 2-3 consecutive errors must be avoided at all costs.** Similarly to quartiles, this will impact your score **more heavily than your total amount of errors.**

You must **be particularly wary of this rule in Q1 and Q2** as their effect will be stronger the earlier on these series of mistakes occur.

3. Never leave any questions unanswered even if short on time, better to guess/select wrong answers

The GMAT algorithm **heavily penalises not answering questions.** Although it might be tempting to try and cram a last thought-out answer on a question when time is nearly out, be sure instead to be able to **finish all questions EVEN if answering them randomly. A random answer is better than no answer at all** - besides your odds of getting a correct answer, although low, remain higher.

4. Improving your GMAT score is NOT about TOTAL accuracy

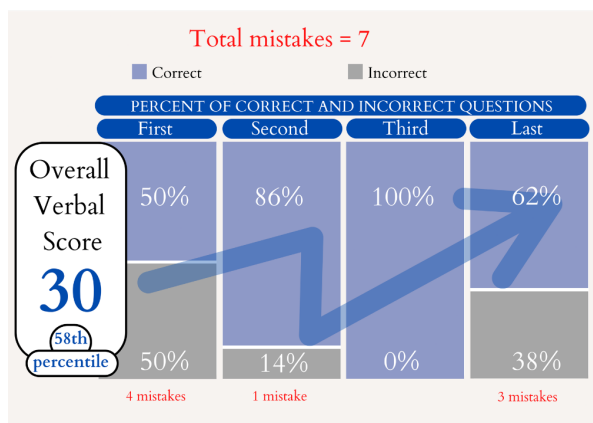
Let us clarify - your GMAT score will be **heavily dependent** upon your **accuracy on Q1 and Q2** rather than your total accuracy. Ultimately, your ability to really perform and cross the 680 threshold will depend on your ability to answer **medium** and **hard questions**, thus increasing your accuracy on Q1 and Q2 sections.

Below you will find visual explanations of the above through an analysis of various **GMAT Enhanced Score Reports (ESR)**



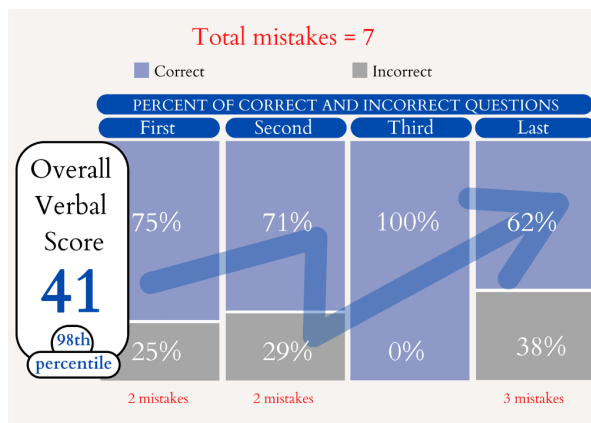
1) The number of mistakes is not representative of the final score

Example 1



This test-taker is 58th percentile

Example 2



This test-taker is 98th percentile

N.B.: That is a difference of approx. 80 points on the GMAT

This goes to show that **where and how** you make your errors on your GMAT is **critical to your final score** - **this is what most students do not understand** and where **your knowledge** of the algorithm can **truly make a difference**.

7
mistakes

93 Percentile

This is V41

8
mistakes

58 Percentile

This is V30

9
mistakes

26 Percentile

This is V21

(This is a visually different representation of the same point made above)



III. Recommended Verbal Strategy (aiming for V40+)

1. Focus on the first 27 questions

The first 27 questions will matter the most - if necessary, solve only 50% of questions in the last quarter

2. Verbal Timing Strategy per question type

Manage your time so that you **never** leave any questions unanswered.

If you are short on time, seek to understand which type of question is draining most of your time and re-work on your **methodology**. Recommended timing per type of question:

- SC: 1m30s
- CR: 2mn
- RC: 1m50s

Feel free to adjust your time **downwards** by 20s on your **strongest areas**, and **upwards** by 20s on your **weakest areas**.

3. Verbal Timing Strategy per Quartile

Per quartile we recommend:

- Q1: 1m40 (easy-medium)
- Q2: 2m (medium)
- Q3: 2m20 (medium-hard)
- Q4: 1m30 (hard)

We have accounted for the level of difficulty typically associated with each quartile here.

IV. Recommended Quant Strategy (aiming for Q47+)

1. Focus on the first 15 questions

The first 15 questions will matter the most here - **take the time necessary to answer them correctly at all costs**.

We repeat this throughout, but **particularly for quant, avoid consecutive mistakes by all means necessary**.

2. Quant Timing Strategy

Per Quartile we recommend:

- Q1: 1m30 (medium / medium hard)
- Q2: 2m10 (medium-hard / hard)
- Q3: 2m40 (hard / very hard)
- Q3: 1m40 (very hard)



3. Accuracy

You must **at all costs** get easy and medium questions right in all areas, including your **weak ones**.

By extension, **do not take seemingly simple questions too lightly** (particularly DS questions) - avoiding making **dumb mistakes** or **falling for traps**.

Being accurate during the test entails devoting the right amount of time per type of question **and per difficulty level in practice** - keep that in mind.

Once you reach Q3, try maintaining decent accuracy; however, once you realise you **are short on time**, solve questions selectively.



V. Improving quickly in your GMAT Prep

1. Focus on finding **METHODICAL** solutions and approaches

Put plainly, focus on **how** you answer your questions, much more than focusing on **why** an answer is right or wrong for a particular question.

- You want to be able to **scale your learning geometrically, not linearly** - it's all about **quality**, not quantity, despite what most candidates think.
- This means that in order to **maximise your improvement** you should **also look at in-depth corrections for your CORRECT answers as well**; this **guarantees you have applied the most methodical solution that you will be able to apply to further questions in different circumstances**.
- Identify your patterns in incorrect responses

Focusing on **METHOD** will help you consistently succeed on medium and hard questions, those most vital to helping you pass the 700 barrier.

2. Reduce your exam anxiety

- Instead of focusing on cramming more and more exercises, focus on **doing fewer questions**, but spending **MORE time working in-depth on the solutions**
- **Strategic time-outs** (we recommend 20sec, find what works for you) during the test when you feel **overwhelmed**/feel as if you are **losing excessive time due to pressure**; this will allow you to recentre, and **shift your mindset for the rest of the test** rather than dragging it out.

3. Truly learning from your mistakes

- **Purchase your GMAT Enhanced Score Report (ESR)** - this investment will be well worth it by helping you identify your **weak modules, concepts**, how you perform in different **quartiles** of the GMAT, your **timing** issues, where taking a strategic 20-sec pause could benefit you, and much more.
 - **Visit:** <https://www.mba.com/exam-prep/gmat-enhanced-score-report>
- The process we suggest for learning from your mistakes:
 - First, cover your **concepts gaps**
 - Then, do sets of 10 to **practice fixing your approach** - 1 at a time, **identify your wastes**
 - Thirdly, practice **solidifying your approaches** - practice sets of 30/40 questions and **work on gradually reducing the time you take**
 - **Do not do only step 3 (or 1 & 3)**
 - **Keep an error log (or use [ours](#))** - some elements to track
 - Primary concept tested



- Secondary concept tested
 - Time taken to solve
 - Accuracy
 - Difficulty level
 - Application-based errors made
 - Source of the question
- When studying, **do NOT skip concepts you think you are half comfortable with** - given Q1 is the most important part of your GMAT and the questions will be random, you **should be able to answer medium questions for all concepts**.
 - **Do not mistake activity for achievement & do not mistake memorization for knowledge.** Many of us have successfully navigated higher education to this point by cramming for tests, grinding out papers, and counting on attendance and participation grades to round out our scores. The GMAT is a different animal, and many students are frustrated by putting in time and effort but not seeing results. To succeed on this test you need to hold yourself accountable for what you learn and not just what you do. **Repetition and memorization have their place, but only if they are building blocks toward deeper conceptual and strategic understanding.**

4. Timing: is your timing issue dependent on specific concepts (divisibility, primes, quadratic inequalities) or modules (take longer on RC)

The art of skipping (guessing) questions:

- As a rule of thumb, keep in mind that your test will follow somewhat of **Pareto's principle**: you will spend **2 minutes on 80% of your questions**, and the **20% will take much more**.

3 types of questions you should guess on the GMAT:

- **Tough Questions:** plainly and simply questions you know are out of your grasp and are not worth spending time on.
- **Tricky Questions:** questions that leave you questioning the required task - i.e a sentence correction question where you end up eliminating all options.
- **Tedious questions:** questions you are capable of solving, but that are too time intensive - i.e a quant question that you know will involve a lot of number crunching to come up with an answer

We are all guilty of spending more time on questions than we should have or failing to recognize quickly enough which questions to skip, but **mastering this underrated skill will without a doubt improve your score.**

However, keep in mind you **SHOULD NOT** be guessing questions in the **Q1 and Q2 portions of the test**, the later on in the test the better.



Making educated vs blind guesses:

Verbal: Sentence Correction (SC)

- **Educated:** The deal with SC questions is you can be stuck in the last two options. If all things fail, you may just want to use your “ear” to pick an answer – just don’t waste more time sitting on the fence by weighing the options in your mind.
- **Blind:** The best questions to guess will be the ones in which the complete sentence is underlined, and the answer options have different sentence structures. These questions typically require you to mentally juggle the sentence structures and ensure you are retaining the meaning of the sentence.

Verbal: Critical Reasoning (CR)

- **Educated:** In CR, you should be able to remove at least a few answer options using common traps such as out-of-scope, extreme answers, and vague phrases.
- **Blind:** If you have not gotten a good sense after reading the stimulus, guess early. Do not get to the answer options and waste more time. There is very little chance you will miraculously get to the correct choice. You will waste time and drain valuable mental energy on the test.

Verbal: Reading Comprehension (RC)

- **Educated:** Inference questions – while reading answer options, beware of each word. On tougher questions, the GMAT will play on the scope by adding just one word. It is vital that you put the last two options under a microscope and understand the correct contextual meaning of the words.
- **Blind:** The best questions to guess are the ones that start with “Which of the following predictive statements the author would agree with”. Especially if you think the passage is very dense and you are not sure where to look for the answer. Remember that we cannot afford to guess all the questions in an entire passage, but we can maximize our ROI by guessing one question among the three / four questions for any given passage.

Quant: Problem Solving (PS)

- **Educated:** If you can get to the equation but are somehow getting lost in actually solving it – ask yourself: what are the values that CANNOT be the answer. This will help eliminate a few, and you can guess what is left.
- **Blind:** If you cannot construct an equation / or know the rule to solve the question, then it might be wise to recognize this early and guess without wasting too much time.



Verbal: Data Sufficiency (DS)

- **Educated:** Typically, the trap is when you see that both the options together will help answer the question, i.e. the answer is C. Just ensure you understand that either of the two statements alone won't get you the answer.
- **Blind:** If this is a concept or a question you have never seen before, then with time pressure, there is a chance that your brain may not be able to process the problem well. It's best to let go of such questions.

5. Focusing on your GMAT approach

1) Wrong approaches that are very popular for GMAT candidates

Working on your **question approach** and **process** will be vital in making you more **time-efficient** and increasing your **accuracy** early on. Here are a few things **you shouldn't do**:

- Read the 1st and last line of every paragraph and jump to questions
- Read the passage quickly and take a mental note of what is where to refer back to after every question
- Read the question first so you know what to look for

Overall wrong process:

- Read question quickly (20-30s)
- Rough understanding of what is going on
- Eliminate 1-3 choices (20-30s)
- Get stuck on remaining choices (20-30s)
- Re-read question (60-90s)
- Make your best guess

Total time: 2-3:30mn

2) The correct approach used by the best-performing GMAT scorers - the "Time-saving irony"

- Read properly the 1st time to limit back and forths
- Make sure you have a crystal clear understanding of what you are seeking to find
- Eliminate choices confidently

Total time: 1-2mn per question

6. Maximising your GMAT prep

When preparing for the GMAT it is extremely tempting to dive straight in and move on quickly from concept to concept once you get a few correct answers. As discussed previously, you should concentrate on METHODOLOGY and practising sets of concept-related questions to solidify your skills, one concept, one module at a time.



For beginners, we recommend following this order for the Verbal and Quant sections:

Order for your Verbal preparation:

The skills tested in the verbal section are **1° comprehension skills**, **2° analysis skills**, and **3° ability to focus on main points**. Given how intertwined such skills are, it is **important to master them interdependently**. To do so, we would **strongly recommend following this order** in your prep:

1. Sentence Correction
2. Critical Reasoning
3. Reading Comprehension

Order for your Quant preparation:

1. Number Basics
2. Fractions and Decimals
3. Estimation Rounding
4. Linear Equations
5. Quadratic Equations
6. Exponents
7. Unitary Method
8. Ratios and Proportions
9. Percentages
10. Even Odd
11. Primes
12. LCM GCD
13. Linear Inequalities
14. Quadratic Inequalities
15. Absolute Inequalities
16. Profit Loss and Discount
17. Simple and Compound Interest
18. Mixtures
19. Time and Work
20. Time and Distance
21. Divisibility and Remainders
22. Units Digit
23. Statistics
24. Lines and Angles
25. Triangles
26. Quadrilaterals and Polygons
27. Circles
28. 3D Solids
29. Sequence and Series
30. Coordinate Geometry
31. Permutation and Combination



32. Probability

You now have all the tools to ace your GMAT, good luck in your business school applications !
Follow us on [MIM Prep](#) for more free content, check out our [website](#), or [book a call](#) with us if you need further guidance !

